MPLS Introduction

Multi-Protocol Label Switching

Presented by:

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MPLS Introduction

- **Motivation**
- MPLS Basics
- Components and Protocols
- Operation
- Advantages and Disadvantages

Motivation

- IP
  - The first defined and used protocol
  - De facto the only protocol for global Internet working

... but there are disadvantages

Motivation (cont.)

- IP Routing disadvantages
  - Connectionless
    - e.g. no QoS
  - Large IP Header
    - At least 20 bytes
  - Routing in Network Layer
    - Slower than Switching
  - Usually designed to obtain shortest path
    - Do not take into account additional metrics

Motivation (cont.)

- ATM
  - connection oriented
    - Supports QoS
  - fast packet switching with fixed length packets (cells)
  - integration of different traffic types (voice, data, video)

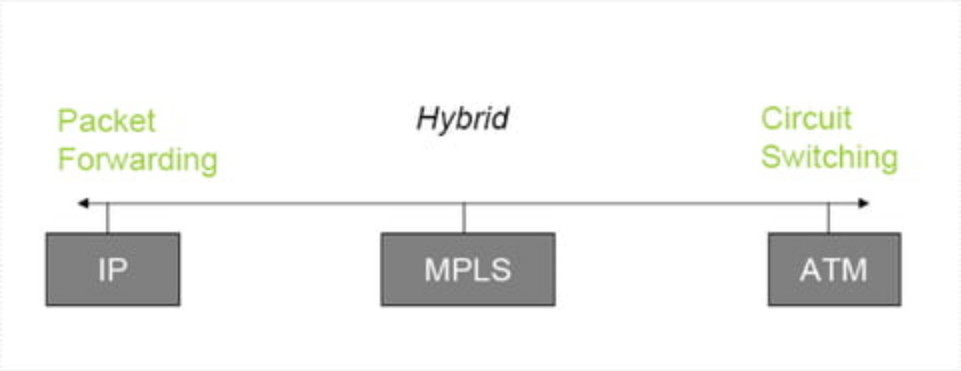
... but there are also disadvantages

Motivation (cont.)

- ATM disadvantages
  - Complex
  - Expensive
  - Not widely adopted

Motivation (cont.)

- Idea: Combine the forwarding algorithm used in ATM with IP.

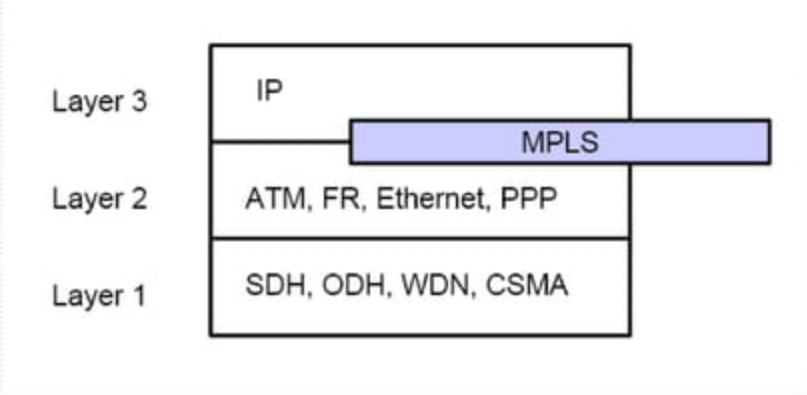


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MPLS Basics

- Multi Protocol Label Switching is arranged between Layer 2 and Layer 3



MPLS Basics (cont.)

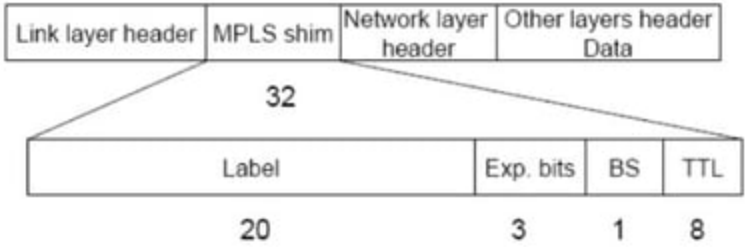
- MPLS Characteristics
  - Mechanisms to manage traffic flows of various granularities (*Flow Management*)
  - Is independent of Layer-2 and Layer-3 protocols
  - Maps IP-addresses to fixed length labels
  - Supports ATM, Frame-Relay and Ethernet

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Label

- Generic label format



Exp.bits:	Experimental Bits, often used for Class of Service
BS:	Bottom of Stack bit, is set if no label follows
TTL:	Time To Leave, used in the same way like in IP

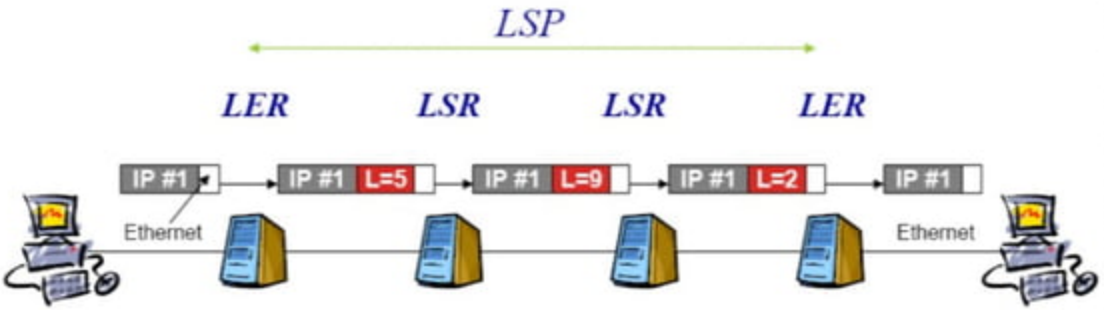
Label Edge Router - LER

- Resides at the edge of an MPLS network and assigns and removes the labels from the packets.
- Support multiple ports connected to dissimilar networks (such as frame relay, ATM, and Ethernet).

Label Switching Router - LSR

- Is a high speed router in the core on an MPLS network.
- ATM switches can be used as LSRs without changing their hardware. Label switching is equivalent to VP/VC switching.

Positions of LERs & LSRs



IP Addr	Out Label
192.4/16	5
Layer 2 Transport	Assign init label

In Label	Out Label
5	9
Label Swapping	

In Label	Out Label
9	2
Label Swapping	

In Label	Next Hop
2	192.4/16
Remove Label	Layer 2 Transport

"ROUTE AT EDGE, SWITCH IN CORE"

Label Distribution Protocol - LDP

- An application layer protocol for the distribution of label binding information to LSRs.
  - It is used to map FECs to labels, which, in turn, create LSPs.
  - LDP sessions are established between LDP peers in the MPLS network (not necessarily adjacent).
  - Sometimes employs OSPF or BGP.

Traffic Engineering

- In MPLS, traffic engineering is inherently provided using explicitly routed paths.
- The LSPs are created independently, specifying different paths that are based on user-defined policies. However, this may require extensive operator intervention.
- RSVP-TE and CR-LDP are two possible approaches to supply dynamic traffic engineering and QoS in MPLS.

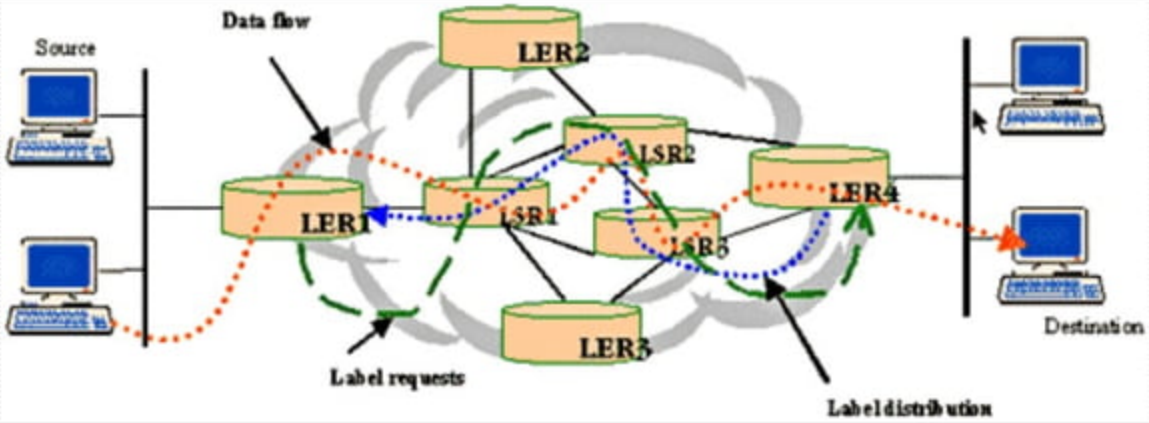
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MPLS Operation

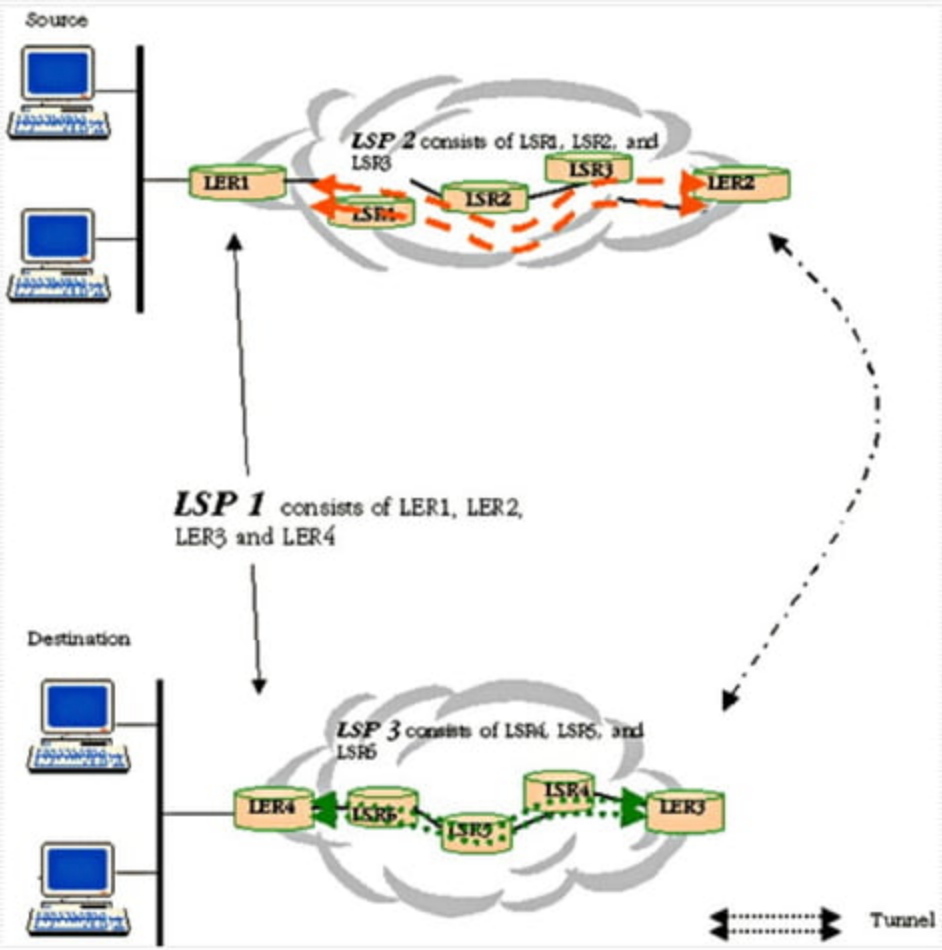
- The following steps must be taken for a data packet to travel through an MPLS domain.
  - label creation and distribution
  - table creation at each router
  - label-switched path creation
  - label insertion/table lookup
  - packet forwarding

MPLS Operation Example



Tunneling in MPLS

- Control the entire path of a packet without explicitly specifying the intermediate routers.
  - Creating tunnels through the intermediary routers that can span multiple segments.
- MPLS based VPNs.


Source
LSP 2 consists of LSR1, LSR2, and LSR3
LER1
LSR1
LSR2
LSR3
LER2
LSP 1 consists of LER1, LER2, LER3 and LER4
Destination
LSP 3 consists of LSR4, LSR5, and LSR6
LER4
LSR6
LSR5
LSR4
LER3
Tunnel

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MPLS Advantages

- Improves packet-forwarding performance in the network
- Supports QoS and CoS for service differentiation
- Supports network scalability
- Integrates IP and ATM in the network
- Builds interoperable networks

MPLS Disadvantages

- An additional layer is added
- The router has to understand MPLS

References

- http://www.iec.org/online/tutorials/mpls/index.html
- http://www.iaik.tu-graz.ac.at/teaching/03_adva
- http://ica1www.epfl.ch/cn2/0304/doc/lecture/m

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Q & A

Thank you!